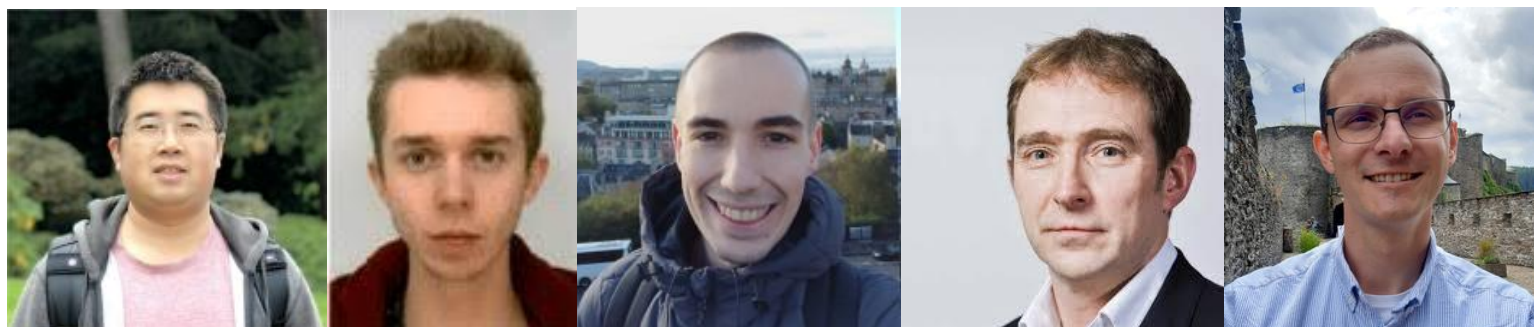


Self-Supervised Learning for Imaging Inverse Problems

Julián Tachella, CNRS, École Normale Supérieure de Lyon



Linear Inverse Problems

Goal: recover signal x from y

$$y = Ax + \epsilon$$

Diagram illustrating the components of the linear inverse problem equation $y = Ax + \epsilon$:

- y : MEASUREMENTS $\in \mathbb{R}^m$
- A : MEASUREMENT OPERATOR $\in \mathbb{R}^{m \times n}$
- x : SIGNAL $\in \mathbb{R}^n$
- ϵ : NOISE $\in \mathbb{R}^m$

Examples

Image denoising

- $A = \text{identity}$

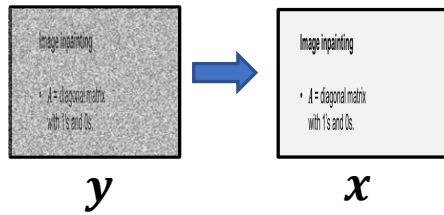
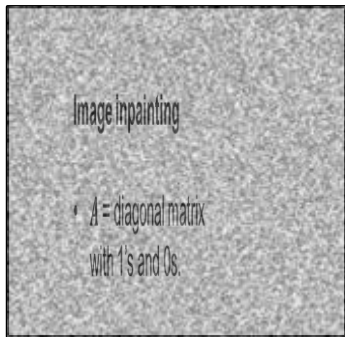
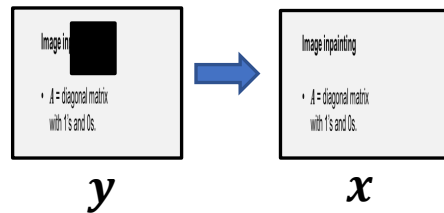
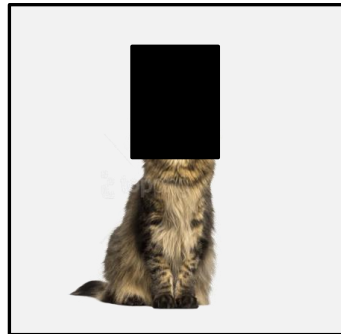


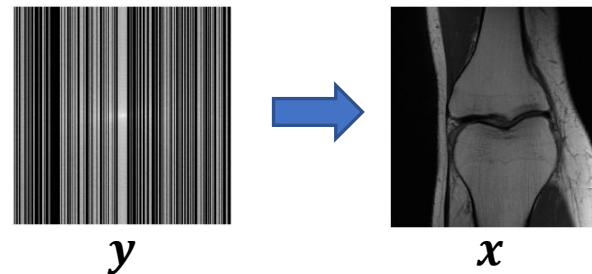
Image inpainting

- $A = \text{diagonal matrix with 1's and 0s.}$



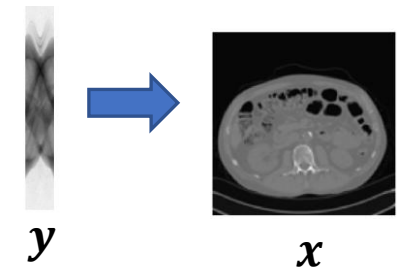
Magnetic resonance imaging

- $A = \text{subset of Fourier modes } (k - \text{space})$



Computed tomography

- $A = \text{1D projections (sinograms)}$



Regularised reconstruction

Idea: define a regularisation $\rho(\mathbf{x})$ promoting plausible reconstructions

$$\hat{\mathbf{x}} = \operatorname{argmin}_{\mathbf{x}} \|\mathbf{y} - A\mathbf{x}\|^2 + \rho(\mathbf{x})$$

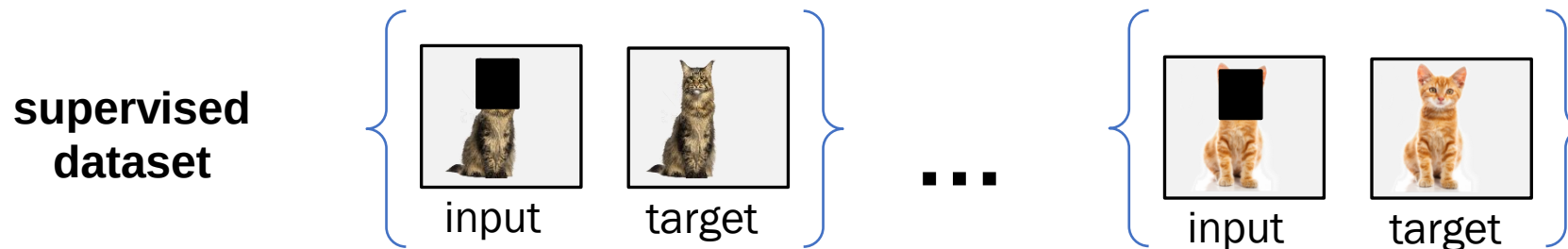
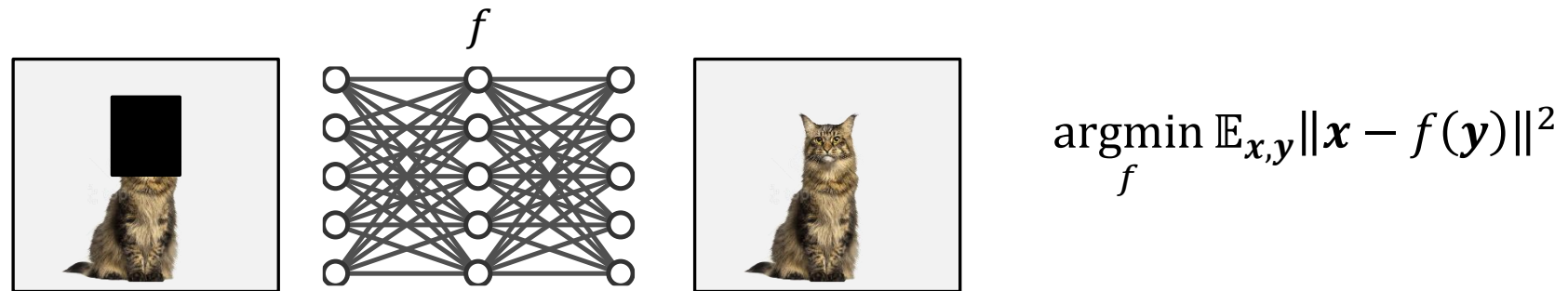
Examples: total-variation, sparsity, etc.

Disadvantages: hard to define a good $\rho(\mathbf{x})$ in real world problems, loose with respect to the true signal distribution



Learning approach

Idea: use training pairs of signals and measurements to directly learn the inversion function



Learning approach

Advantages:

- State-of-the-art reconstructions
- Once trained, f is easy to evaluate

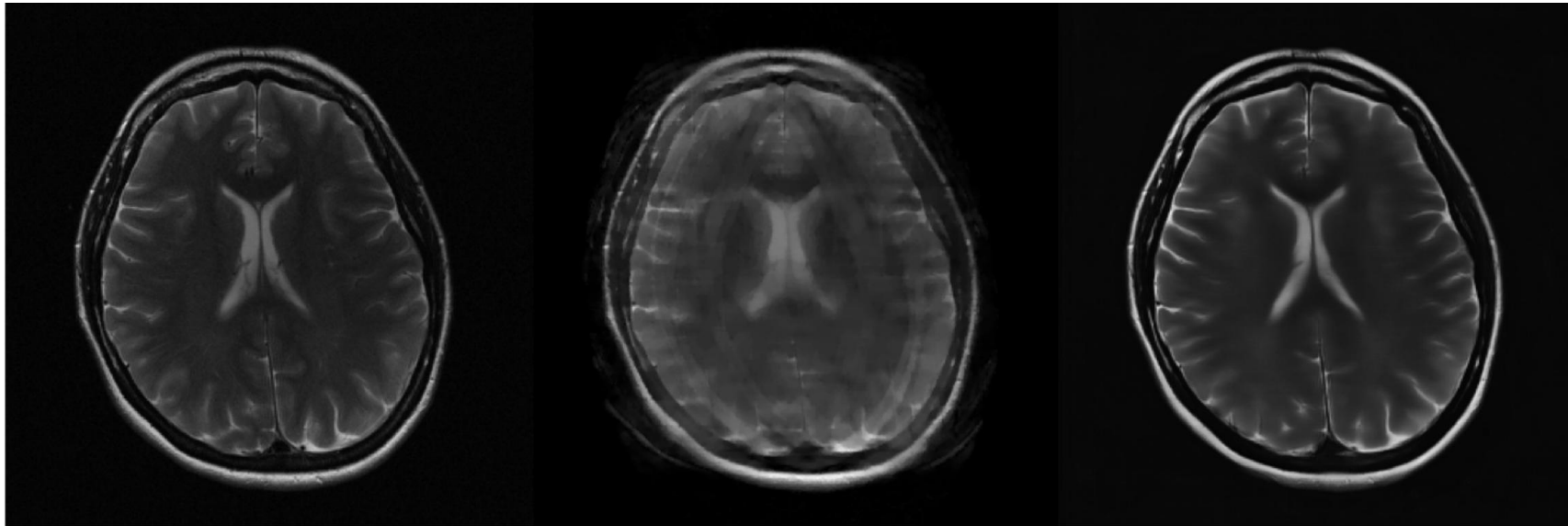
fastMRI

Accelerating MR Imaging with AI

Ground-truth

Total variation
(28.2 dB)

Deep network
(**34.5 dB**)

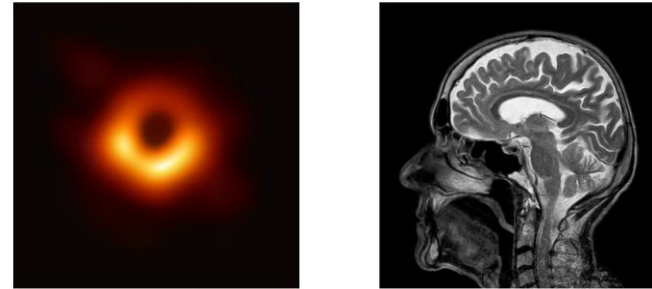


x8 accelerated MRI [Zbontar et al., 2019]

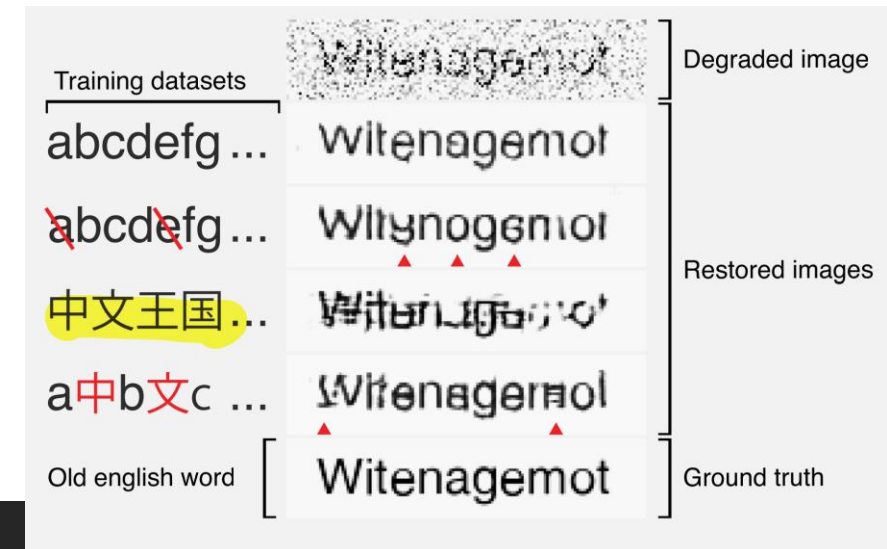
Learning approach

Main disadvantage: Obtaining training signals x_i can be expensive or impossible.

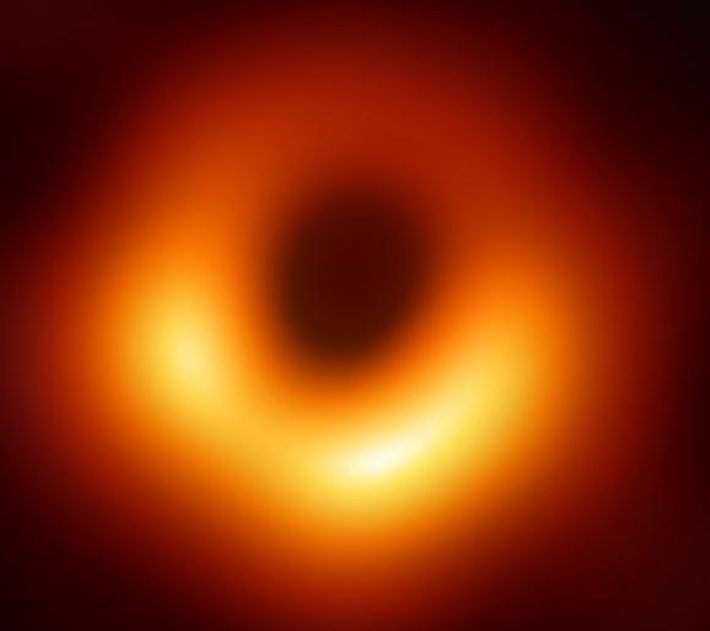
- Medical and scientific imaging



- Distribution shift [Belthangady & Royer, 2019]

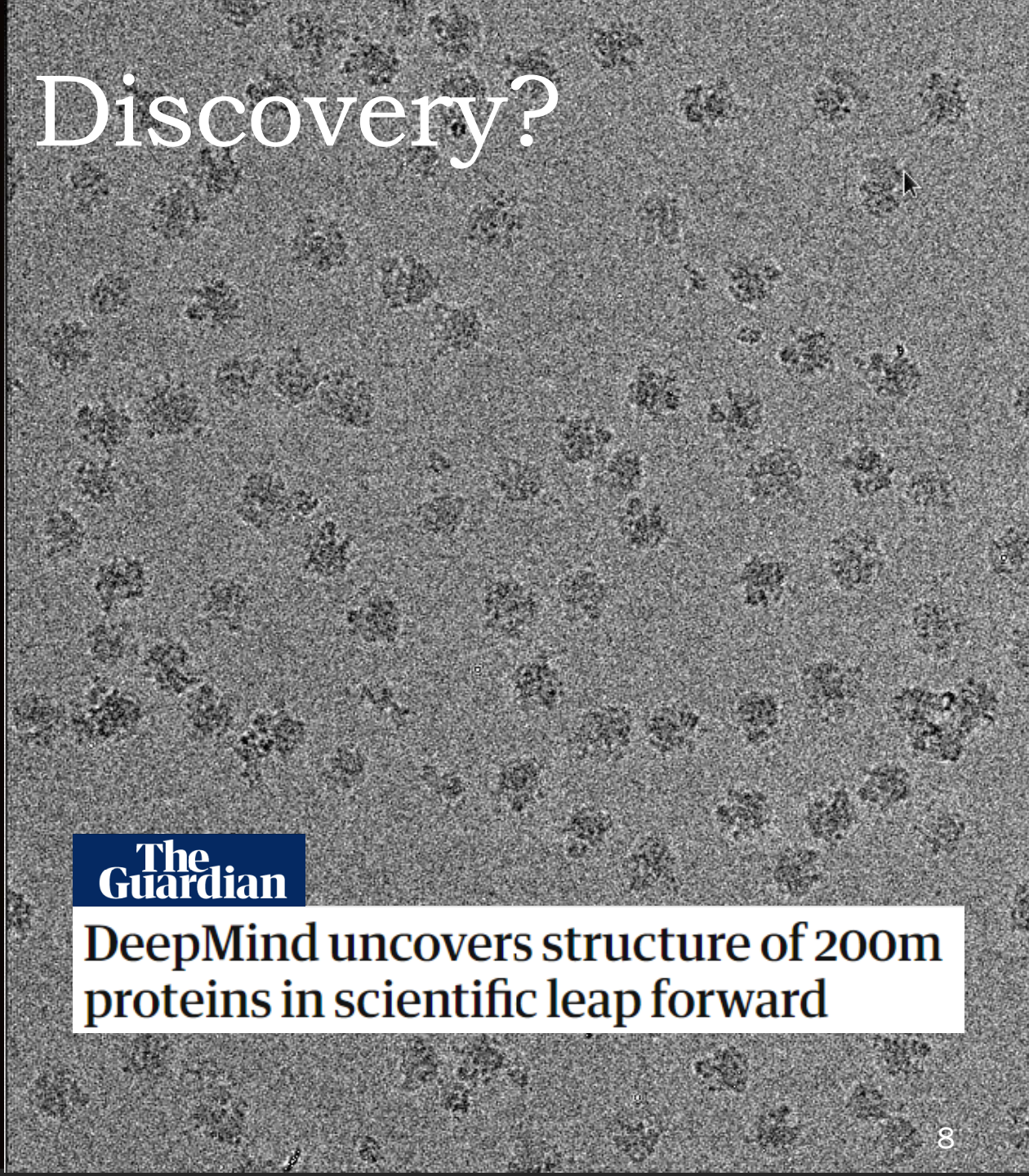


AI for Knowledge Discovery?



The
Guardian

Black hole picture captured for first time in space breakthrough



The
Guardian

DeepMind uncovers structure of 200m proteins in scientific leap forward

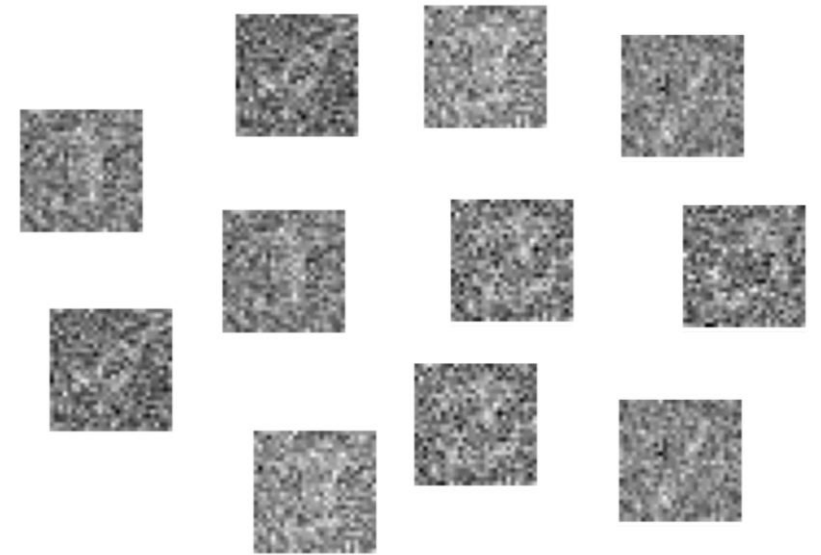
Outline

How can we learn f from measurement $\{\mathbf{y}_i\}$ data alone?

1. Noisy: $\mathbf{y} = \mathbf{x} + \boldsymbol{\epsilon}$
2. Incomplete and noisy: $\mathbf{y} = A\mathbf{x} + \boldsymbol{\epsilon}$

Goal: build a self-supervised loss $\mathcal{L}_{\text{SELF}}$ such that

$$\mathbb{E}_{\mathbf{y}} \mathcal{L}_{\text{SELF}}(\mathbf{y}, f) = \mathbb{E}_{\mathbf{x}, \mathbf{y}} \mathcal{L}_{\text{SUP}}(\mathbf{x}, \mathbf{y}, f) + \text{const.}$$

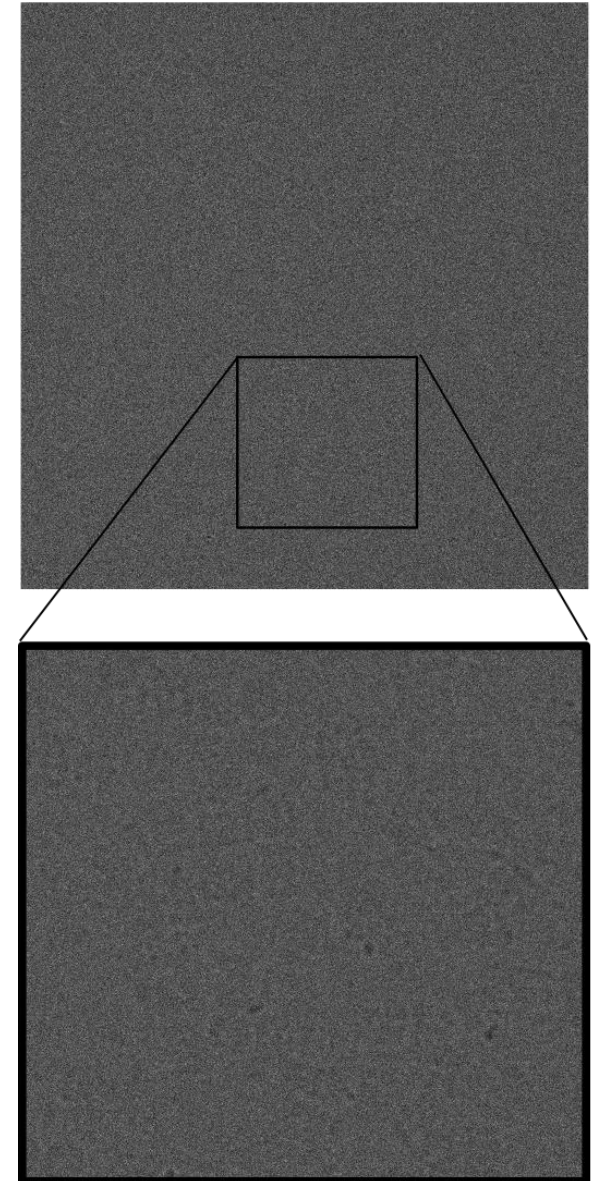


Measurement

How can we learn f from measurement $\{y_i\}$ data alone?

Example

- Cryo-Electron microscopy images
- Extremely low SNR
- Noise distribution is unknown



Self-Supervised Risk Estimators

Goal: build self-supervised loss $\mathcal{L}_{\text{SELF}}$:

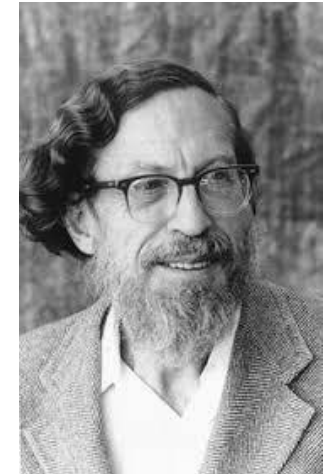
- Unbiased estimator: $\mathbb{E}_{\mathbf{y}} \mathcal{L}_{\text{SELF}}(\mathbf{y}, f) \propto \mathbb{E}_{\mathbf{x}, \mathbf{y}} \|\mathbf{x} - f(\mathbf{y})\|^2 = \mathbb{E}\{\mathbf{x}|\mathbf{y}\}$
- Same global minimum: $\underset{f}{\operatorname{argmin}} \mathbb{E}_{\mathbf{y}} \mathcal{L}_{\text{SELF}}(\mathbf{y}, f) = \underset{f}{\operatorname{argmin}} \mathbb{E}_{\mathbf{x}, \mathbf{y}} \|\mathbf{x} - f(\mathbf{y})\|^2 = \mathbb{E}\{\mathbf{x}|\mathbf{y}\}$
- Same minima **constrained set**: $\underset{f \in \mathcal{F}}{\operatorname{argmin}} \mathbb{E}_{\mathbf{y}} \mathcal{L}_{\text{SELF}}(\mathbf{y}, f) = \underset{f \in \mathcal{F}}{\operatorname{argmin}} \mathbb{E}_{\mathbf{x}, \mathbf{y}} \|\mathbf{x} - f(\mathbf{y})\|^2 \neq \mathbb{E}\{\mathbf{x}|\mathbf{y}\}$

What is the best we can do? **MMSE** estimator

Stein's Unbiased Risk Estimator

$$\mathcal{L}_{\text{SURE}}(\mathbf{y}, f) = \underbrace{\|\mathbf{y} - f(\mathbf{y})\|^2}_{\text{Measurement consistency}} + 2\sigma^2 \underbrace{\sum_i \frac{\delta f_i}{\delta y_i}(\mathbf{y})}_{\text{Degrees of freedom [Efron, 2004]}}$$

Measurement consistency Degrees of freedom [Efron, 2004]



- **Stein's lemma:** Let $\mathbf{y}|\mathbf{x} \sim \mathcal{N}(\mathbf{x}, I\sigma^2)$, then $\mathbb{E}_{\mathbf{y}} \mathcal{L}_{\text{SURE}}(\mathbf{y}, f) \propto \mathbb{E}_{\mathbf{x}, \mathbf{y}} \|\mathbf{x} - f(\mathbf{y})\|^2$
- Extensions for Poisson, Poisson-Gaussian [Hudson, 1978],
- MMSE estimator $f^*(\mathbf{y}) = \mathbb{E}\{\mathbf{x}|\mathbf{y}\}$

Efficient SURE

1) Monte Carlo SURE: approx. divergence as [Ramani et al., 2007]

$$\sum_i \frac{\delta f_i}{\delta y_i}(\mathbf{y}) \approx \frac{\boldsymbol{\omega}^\top}{\alpha} (f(\mathbf{y}) - f(\mathbf{y} + \boldsymbol{\omega}\alpha))$$

$$\boldsymbol{\omega} \sim \mathcal{N}(\mathbf{0}, I\sigma^2) \\ \text{and } \alpha \in \mathbb{R}$$

2) Autodiff SURE: use auto-diff [Soltanayev, 2020]

$$\sum_i \frac{\delta f_i}{\delta y_i}(\mathbf{y}) \approx \boldsymbol{\omega}^\top \left(\frac{\delta f}{\delta \mathbf{y}} \boldsymbol{\omega} \right)$$

3) Recorrup2Recorrup2 [Pang et al., 2021], [Monroy, Bacca & T., CVPR 2025]

$$\mathcal{L}_{\text{R2R}}(\mathbf{y}, f, \alpha) = \mathbb{E}_{\boldsymbol{\omega}} ||\mathbf{y} + \alpha\boldsymbol{\omega} - f(\mathbf{y} - \boldsymbol{\omega}/\alpha) ||^2$$

Extensions to **Poisson, Gamma and Binomial**

Tweedie's Formula



The solution to SURE is **Tweedie's Formula**

$$\min_f \mathbb{E}_{\mathbf{y}} \|\mathbf{y} - f(\mathbf{y})\|^2 + 2\sigma^2 \sum_i \frac{\delta f_i}{\delta y_i}(\mathbf{y})$$

$$\min_f \mathbb{E}_{\mathbf{y}} \|\mathbf{y} - f(\mathbf{y})\|^2 - 2\sigma^2 \sum_i f_i(\mathbf{y}) \frac{\delta \log p_{\mathbf{y}}(\mathbf{y})}{\delta y_i}$$

$$\min_f \mathbb{E}_{\mathbf{y}} \|\mathbf{y} - f(\mathbf{y}) - \sigma^2 \nabla \log p_{\mathbf{y}}(\mathbf{y})\|^2$$



Integration by parts



Complete squares

$$\Rightarrow f(\mathbf{y}) = \mathbf{y} + \sigma^2 \nabla \log p_{\mathbf{y}}(\mathbf{y})$$

- **Noise2Score** [Kim and Ye, 2021] learns $\nabla \log p_{\mathbf{y}}(\mathbf{y})$ from noisy data + denoises with Tweedie.
- Key formula behind diffusion models, which can be trained self-supervised [Daras et al., 2024]

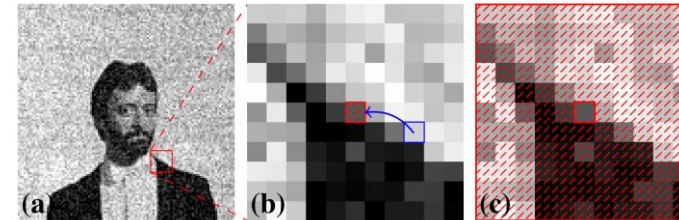
Cross-Validation Methods

What happens if we only know that $p(\mathbf{y}|\mathbf{x}) = \prod p(y_i|x_i)$?

Idea: if f_i doesn't depend on y_i we cannot overfit the noise!

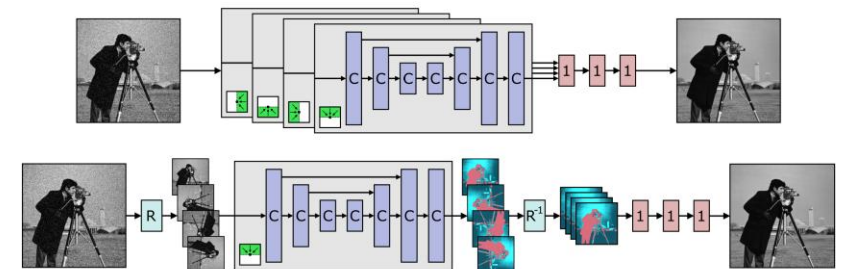
Noise2Void [Krull et al., 2019], **Noise2Self** [Batson, 2019], **Neighbor2Neighbor** [Huang, 2023]

- During training flip centre pixel
- Computes loss only on flipped pixels



Blind spot networks [Laine et al., 2019]

- Convolutional architecture that doesn't 'see' centre pixel by construction



Cross-Validation Methods

- We can write these losses as

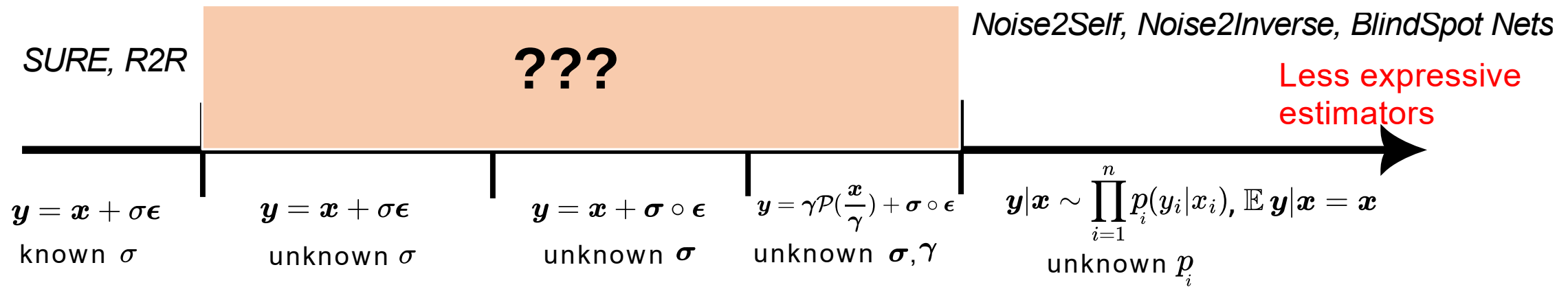
$$\min_f ||\mathbf{y} - f(\mathbf{y})||^2 \text{ subject to } \frac{\delta f_i}{\delta y_i}(\mathbf{y}) = 0 \forall i, \forall \mathbf{y}$$

- SURE's perspective:

$$\min_f ||\mathbf{y} - f(\mathbf{y})||^2 + 2\sigma^2 \sum_i \frac{\delta f_i}{\delta y_i}(\mathbf{y})$$

- These methods are not MMSE optimal!

Are We Missing Something?



UNSURE

- SURE's perspective:

$$\mathcal{L}_{\text{UNSURE}}(\mathbf{y}, f) = \|\mathbf{y} - f(\mathbf{y})\|^2 + 2\sigma^2 \sum_i \frac{\delta f_i}{\delta y_i}(\mathbf{y})$$

- Impose zero-expected divergence [Tachella et al., ICLR25]

$$\mathcal{L}_{\text{UNSURE}}(\mathbf{y}, f) = \|\mathbf{y} - f(\mathbf{y})\|^2 \text{ subject to } \mathbb{E}_{\mathbf{y}} \sum_i \frac{\delta f_i}{\delta y_i}(\mathbf{y}) = 0$$

- In practice, we use Lagrange multipliers

$$\min_f \max_{\eta} \mathbb{E}_{\mathbf{y}} \|\mathbf{y} - f(\mathbf{y})\|^2 + 2\eta \sum_i \frac{\delta f_i}{\delta y_i}(\mathbf{y})$$

UNSURE

- Closed-form solution (similar to Tweedie's formula)

$$\min_f \max_{\eta} \mathbb{E}_{\mathbf{y}} ||\mathbf{y} - f(\mathbf{y})||^2 + 2\eta \sum_i \frac{\delta f_i}{\delta y_i}(\mathbf{y})$$

$$\Rightarrow f^{\text{ZED}}(\mathbf{y}) = \mathbf{y} + \hat{\eta} \nabla \log p_{\mathbf{y}}(\mathbf{y}) \quad \hat{\eta} = \left(\frac{1}{n} \mathbb{E}_{\mathbf{y}} ||\nabla \log p_{\mathbf{y}}(\mathbf{y})||^2 \right)^{-1}$$

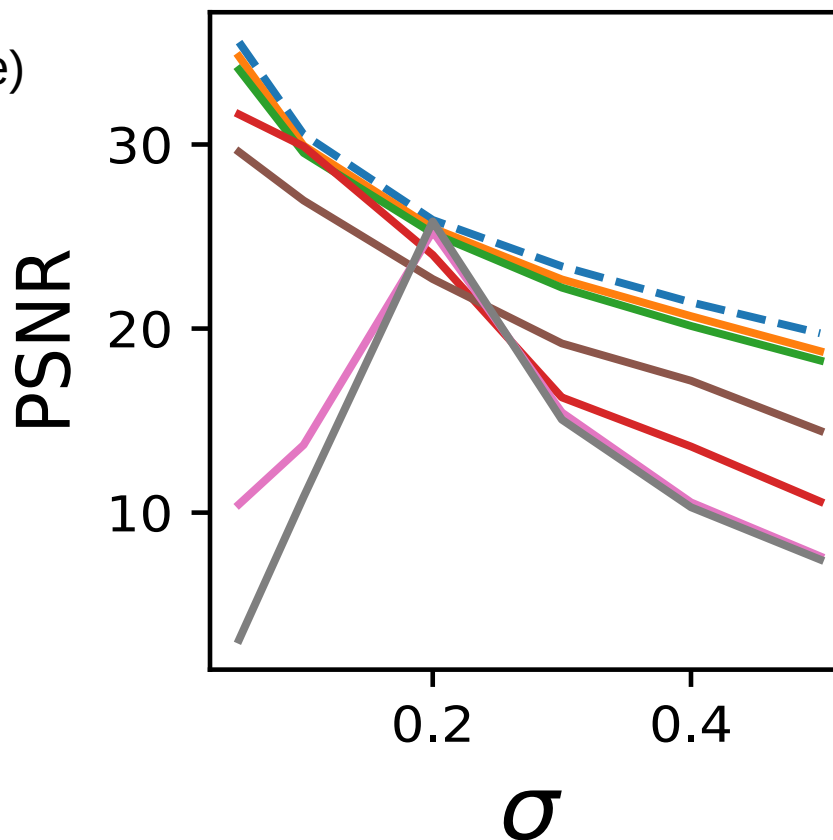
- Expected error

$$\frac{1}{n} \mathbb{E}_{x, \mathbf{y}} ||f^{\text{ZED}}(\mathbf{y}) - \mathbf{x}|| = \sigma^2 \left(\frac{1}{1 - \frac{\text{MMSE}}{\sigma^2}} - 1 \right) \approx \text{MMSE} + \frac{\text{MMSE}^2}{\sigma^2}$$

- UNSURE can be extended to **unknown noise covariance** and **Poisson Gaussian noise**

Denoising Experiments

- MNIST dataset (28x28 grayscale)
- Isotropic Gaussian noise

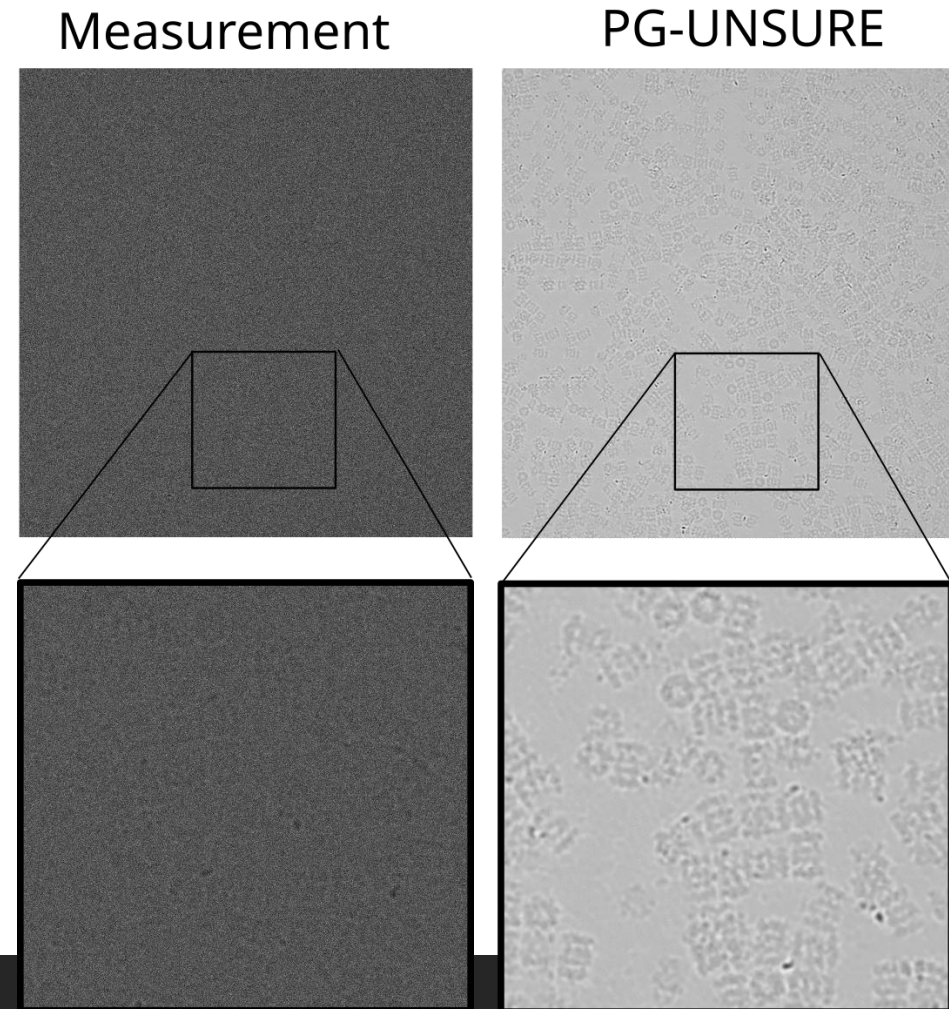


--- supervised — unsure — unsure via score
— neighbor2neighbor — R2R $\sigma = 0.2$ — SURE $\sigma = 0.2$

Poisson-Gaussian UNSURE

Real data experiments

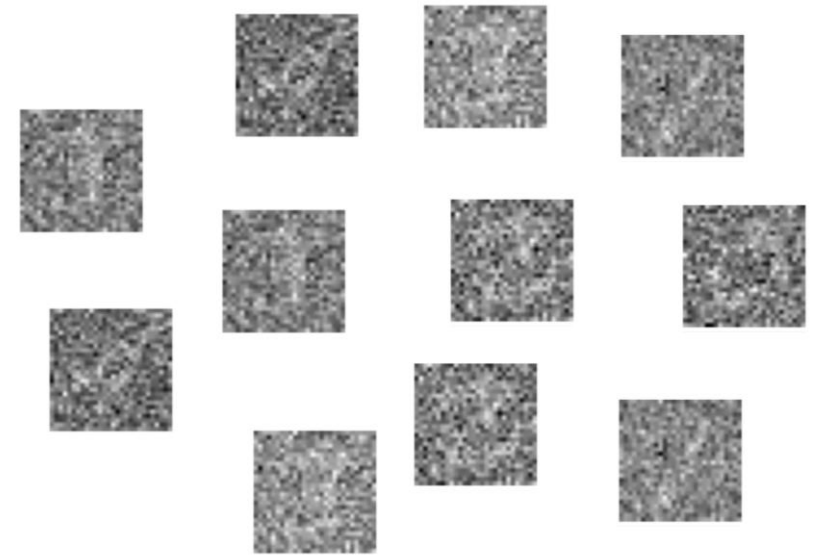
- Cryo electron microscopy images
- Extremely low SNR
- Approx. Poisson-Gaussian noise



Outline

How can we learn f from measurement $\{y_i\}$ data alone?

1. Noisy: $y = x + \epsilon$
2. Incomplete and noisy: $y = Ax + \epsilon$

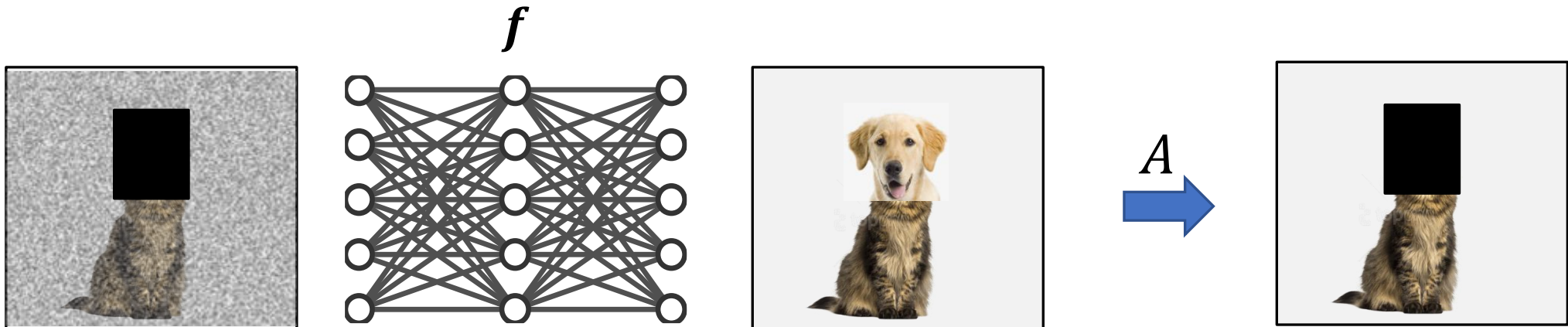


Incomplete Measurements?

For $A \neq I$, most estimators can be adapted to approximate

$$\mathbb{E}_{x,y} ||A(x - f(y))||^2$$

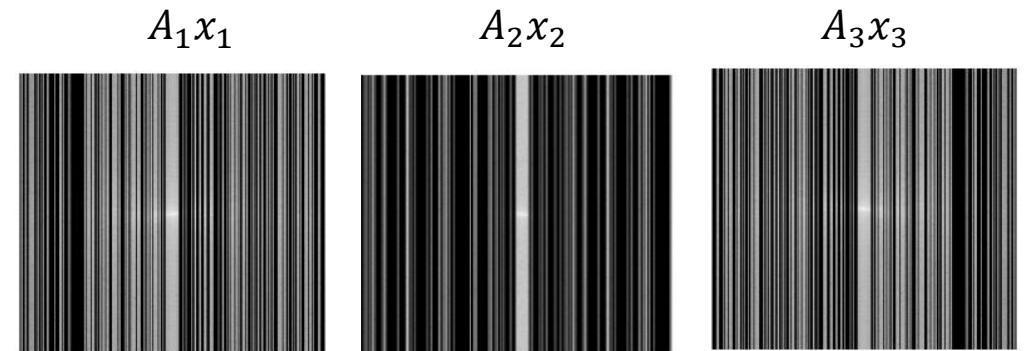
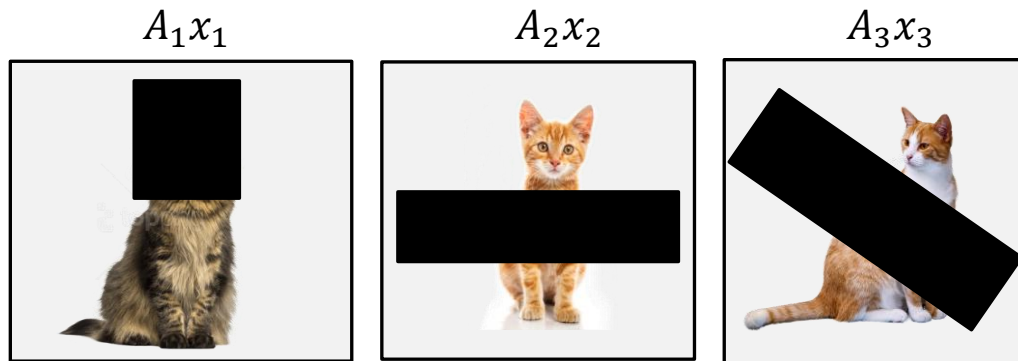
In this case, the risk does not penalise $f(y)$ in the **nullspace** of A !



Learning from Measurements

How to learn from only y ?

- Access multiple operators $y_i = A_{g_i}x_i$ with $g \in \{1, \dots, G\}$
- Each A_g with different nullspace
- Offers the possibility for learning using multiple measurement operators



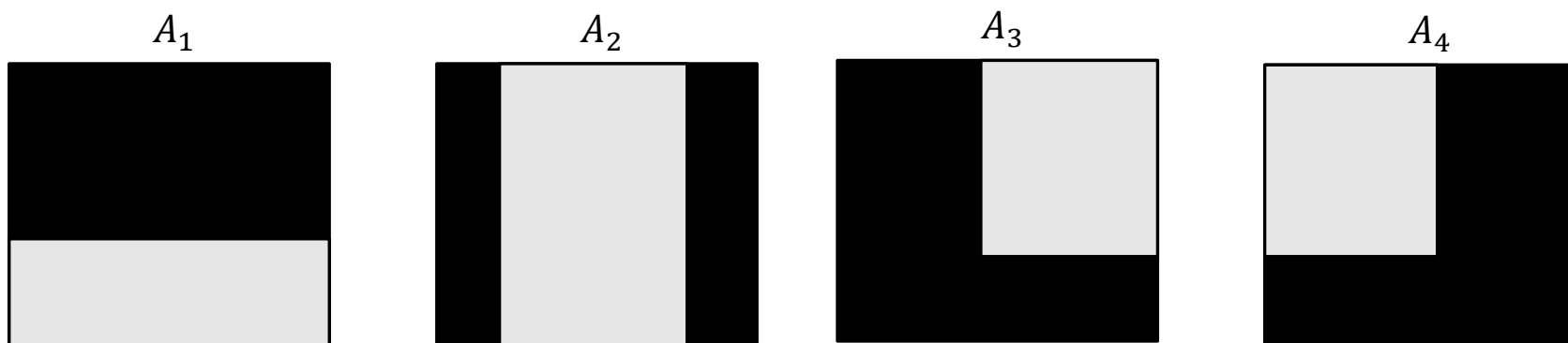
Necessary Condition

Intuition: we need that the operators $A_1, A_2, \dots, A_G$ cover the whole ambient space [Tachella et al., 2023, JMLR].

Proposition: Learning reconstruction mapping f from observed measurements possible only if

$$\text{rank}(\mathbb{E}_g A_g^\top A_g) = n$$

and thus, if $m \geq n/G$.



Multi Operator Imaging

Multi Operator Imaging (MOI) [Tachella et al., 2022, NeurIPS]

$$\mathcal{L}_{\text{MOI}}(\mathbf{y}, f) = \underbrace{\|\mathbf{y} - A_g f(\mathbf{y}, A_g)\|^2}_{\text{Can be replaced by SURE, UNSURE, etc.}} + \sum_s \underbrace{\|f(A_s \hat{\mathbf{x}}, A_s) - \hat{\mathbf{x}}\|^2}_{\text{Enforces } f(A_g \mathbf{x}, A_g) \approx f(A_s \mathbf{x}, A_s)} \quad \text{with } \hat{\mathbf{x}} = f(\mathbf{y}, A_g)$$

Can be replaced by SURE,
UNSURE, etc.

Enforces $f(A_g \mathbf{x}, A_g) \approx f(A_s \mathbf{x}, A_s)$

Inpainting Experiments

- U-Net network
- CelebA dataset
- $G = 40$ different A_g inpainting masks

Measurements y



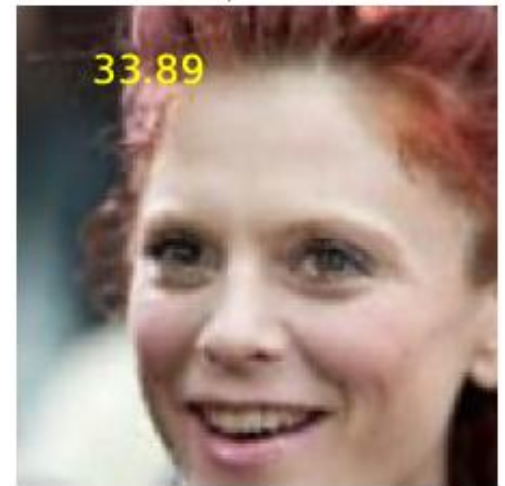
Signal x



Supervised



MOI



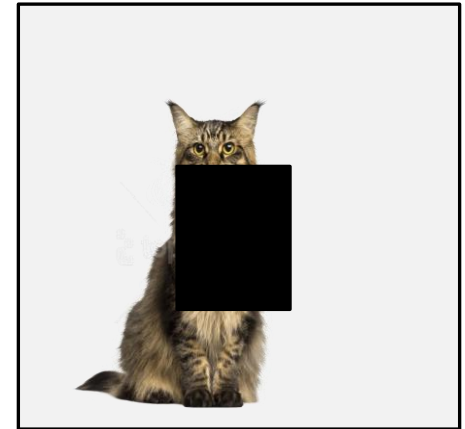
Symmetry Prior

Equivariant Imaging [Chen, Tachella and Davies, ICCV 2021]

For all $g \in G$ we have

$$\mathbf{y} = A\mathbf{x} = \underbrace{AT_g}_{A_g} \overbrace{T_g^{-1}\mathbf{x}}^{\mathbf{x}'} = A_g\mathbf{x}'$$

- We get multiple virtual operators $\{A_g\}_{g \in G}$ 'for free'!
- Each AT_g might have a different nullspace



(Non)-Equivariant Operators

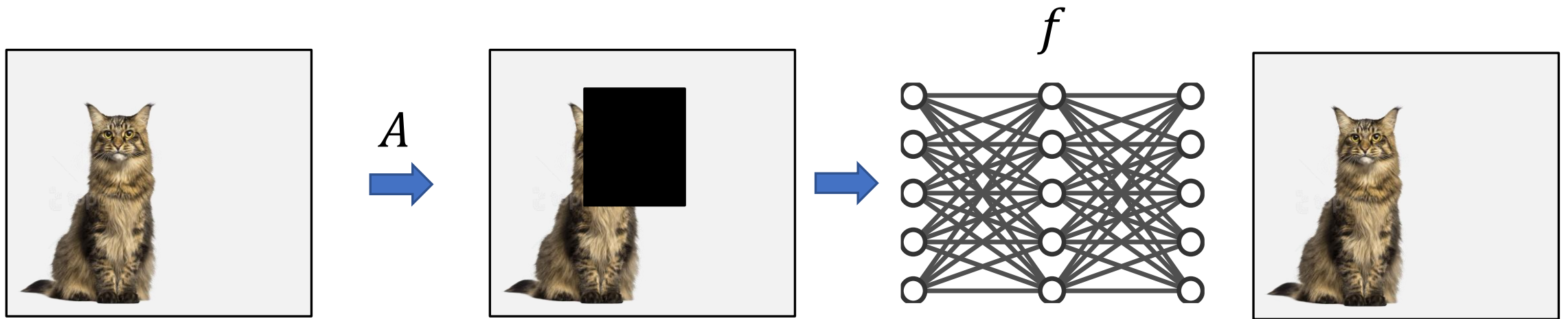
Theorem [T. et al., 2023]: The full rank condition requires that A is **not equivariant**: $AA^T T_g \neq T_g AA^T$

$$\text{rank}(\mathbb{E}_g T_g^T A^T A T_g) = \text{rank}(A^T (\mathbb{E}_g \tilde{T}_g^T \tilde{T}_g) A) = \text{rank}(A^T A) = m < n$$

Equivariant Imaging

How can we enforce equivariance in practice?

Idea: we should have $f(AT_g\mathbf{x}) = T_gf(A\mathbf{x})$, i.e. $f \circ A$ should be G -equivariant



Equivariant Imaging

We can leverage invariance of $p(\mathbf{x})$ to transformations T_g to learn in the nullspace

$$\mathcal{L}_{EI}(\mathbf{y}, f) = \mathbb{E}_g || T_g \hat{\mathbf{x}} - f(AT_g \hat{\mathbf{x}}) ||^2$$

where $\hat{\mathbf{x}} = f(\mathbf{y})$ is used as reference

Robust Equivariant Imaging++ [Chen, Tachella & Davies 2022, CVPR]

enforces equivariance of $f \circ A$

$$\mathcal{L}_{REI}(\mathbf{y}, f) = \underbrace{\mathcal{L}_{UNSURE}(\mathbf{y}, f)}_{\text{Handles noisy measurements of unknown noise level}} + \underbrace{\mathcal{L}_{EI}(\mathbf{y}, f)}_{\text{enforces equivariance of } f \circ A}$$

Handles noisy measurements of unknown noise level

Experiments

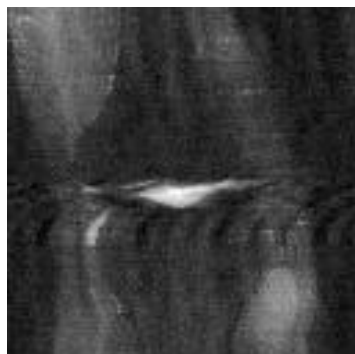
- FastMRI dataset
- Gaussian noise

Method	CV + EI	EI (assumes $\sigma = 0$)	UNSURE + EI (unknown σ)	SURE + EI (assumes $\sigma = 0.05$)	Supervised
PSNR [dB]	33.25 ± 1.14	34.32 ± 0.91	35.73 ± 1.45	28.05 ± 4.73	36.63 ± 1.38

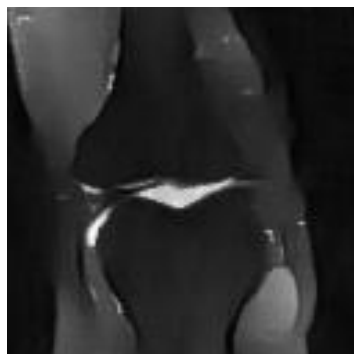
Ground-truth



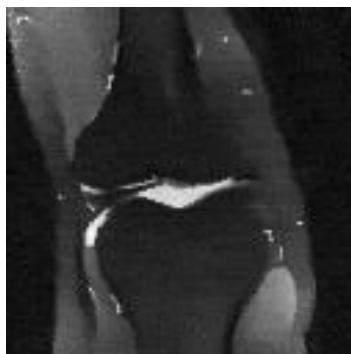
Measurement



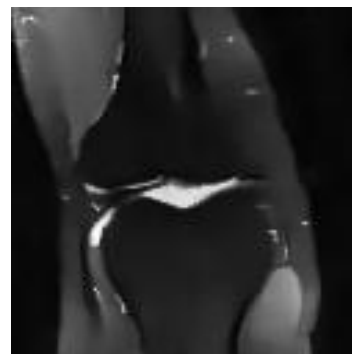
Noise2Inverse



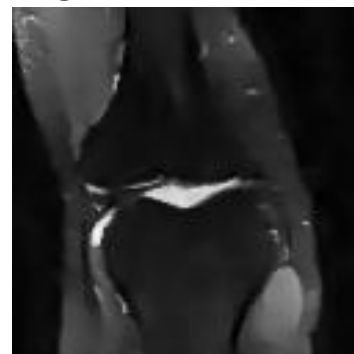
EI $\sigma = 0$



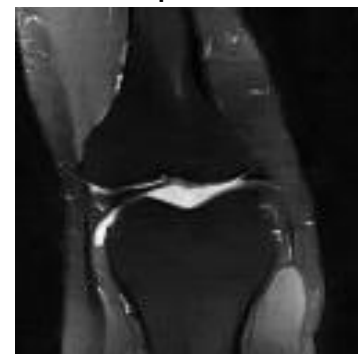
UNSURE+EI



SURE+EI $\sigma = 0.05$

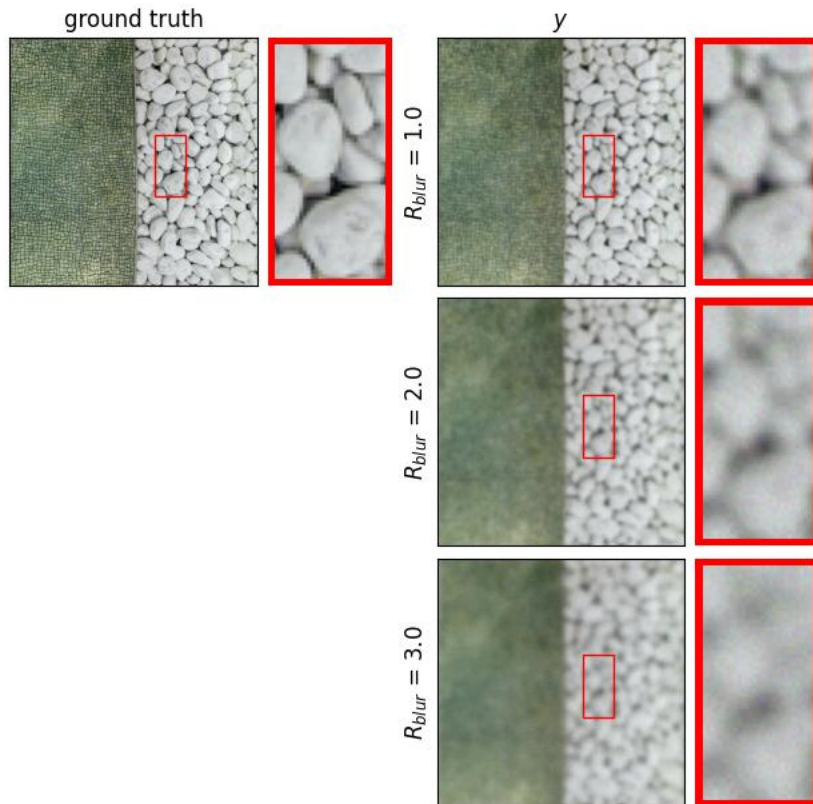


Supervised



Experiments

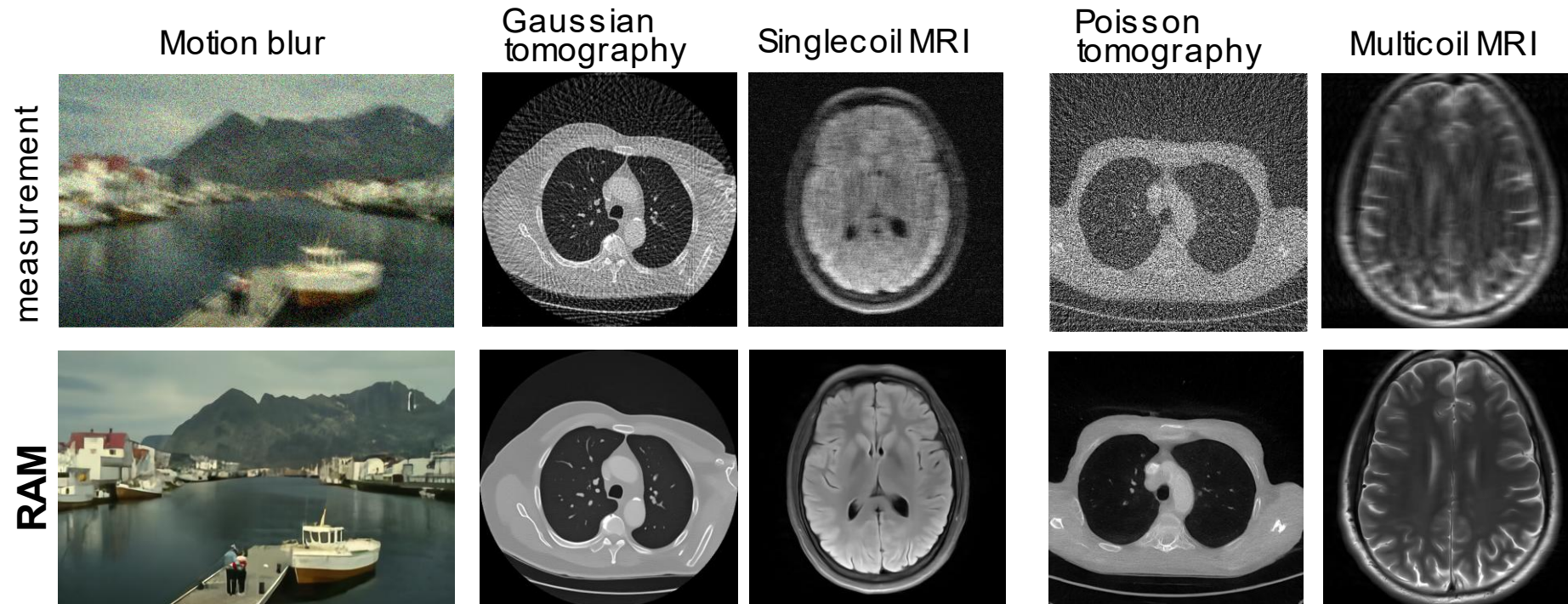
- Operator A is isotropic blur with Gaussian noise
- Dataset is approximately **scale invariant**



Scanvic, Davies, Abry, T., *arxiv* 2024

Finetuning

- **Reconstruct Anything Model:** general model solving many inverse problems [Terris et al., 2025]



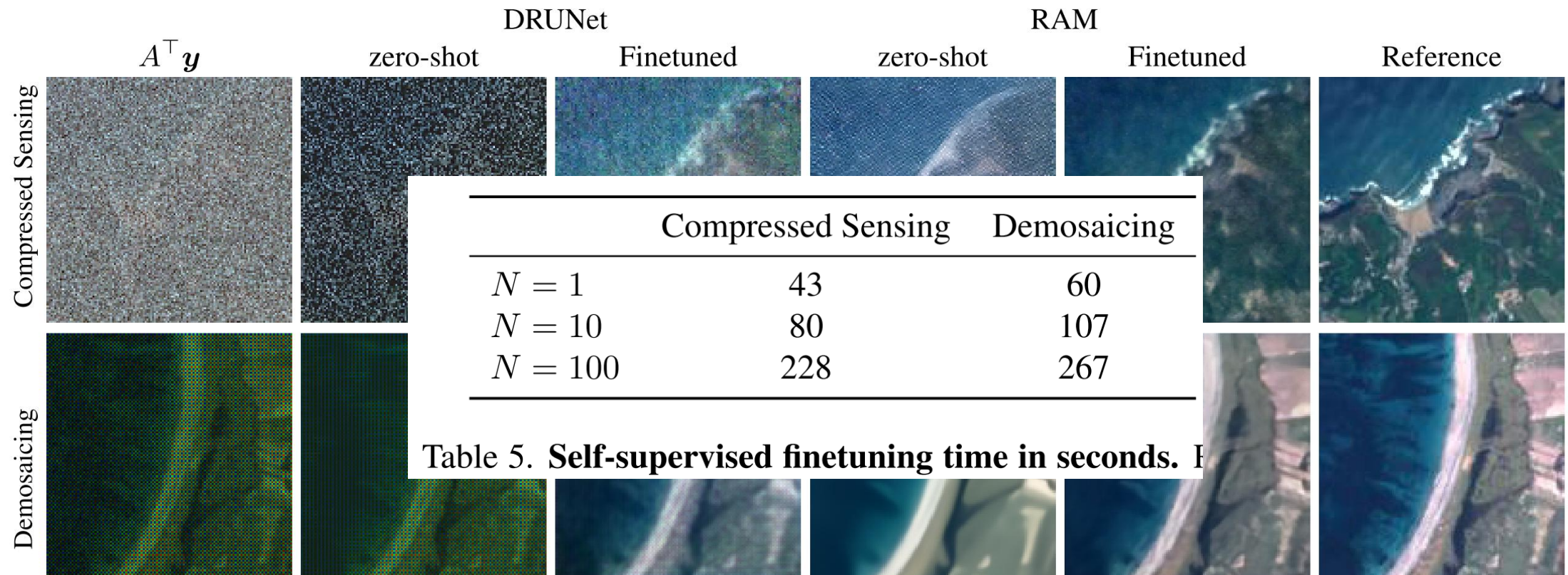
In-distribution degradation and images

Out-of-distribution degradation

Zero-shot performance

Finetuning

- The model can be finetuned with self-supervised losses on up to a single $\mathbf{y}$ ($N = 1$)
- Finetuning can be done in a **few seconds**



References

Slides and codes of a recent 3-hour tutorial can be found here:

<https://tachella.github.io/blog/selfsuptutorial/>





[Quickstart](#) [Examples](#) [User Guide](#) [API](#) [Finding Help](#) [More](#) ▼

Section Navigation

- Basics ▼
- Optimization ▼
- Plug-and-Play ▼
- Sampling ▼
- Unfolded ▼
- Patch Priors ▼
- Self-Supervised Learning ▼
- Adversarial Learning ▼
- Advanced ▼

🏠 > Examples

Examples

All the examples have a download link at the end. You can load the example's notebook on [Google Colab](#) and run them by adding the line

```
pip install git+https://github.com/deepinv/deepinv.git#egg=deepinv
```

to the top of the notebook (e.g., [as in here](#)).

Basics



measurement ground truth DP

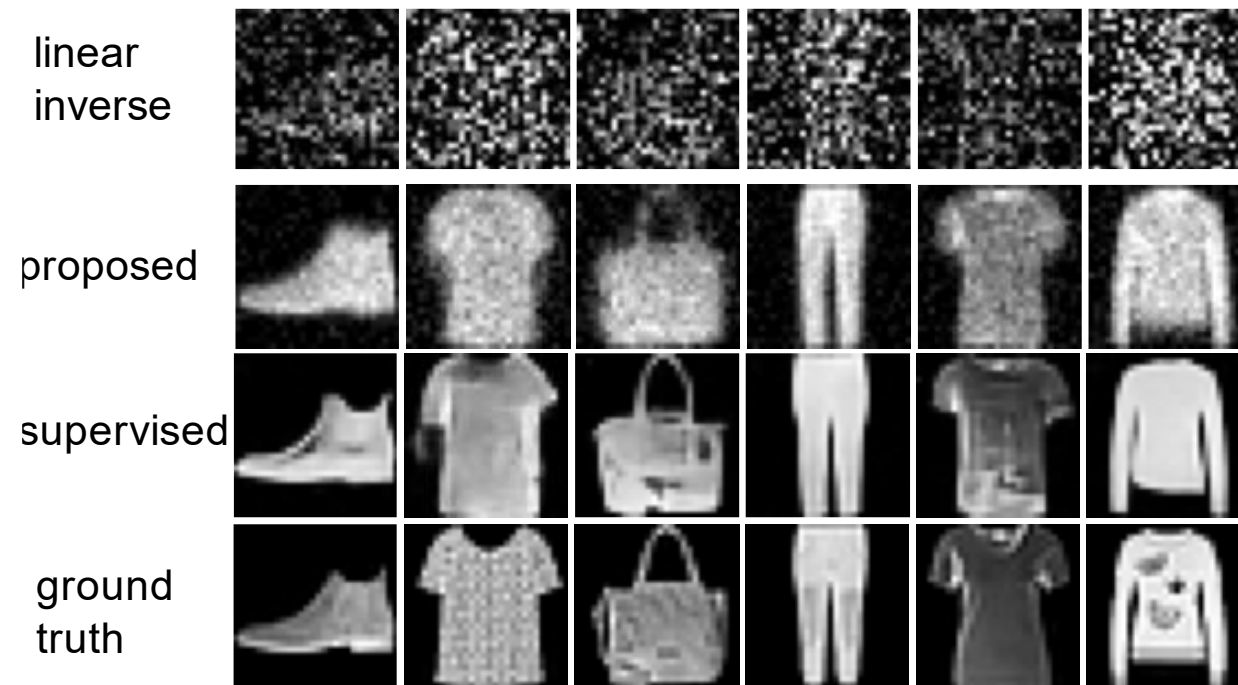


Deep Inverse



1-bit Compressed Sensing

- $y = \text{sign}(Ax)$ with Gaussian A and 20% undersampling ratio
- Dataset is approximately **translation invariant**



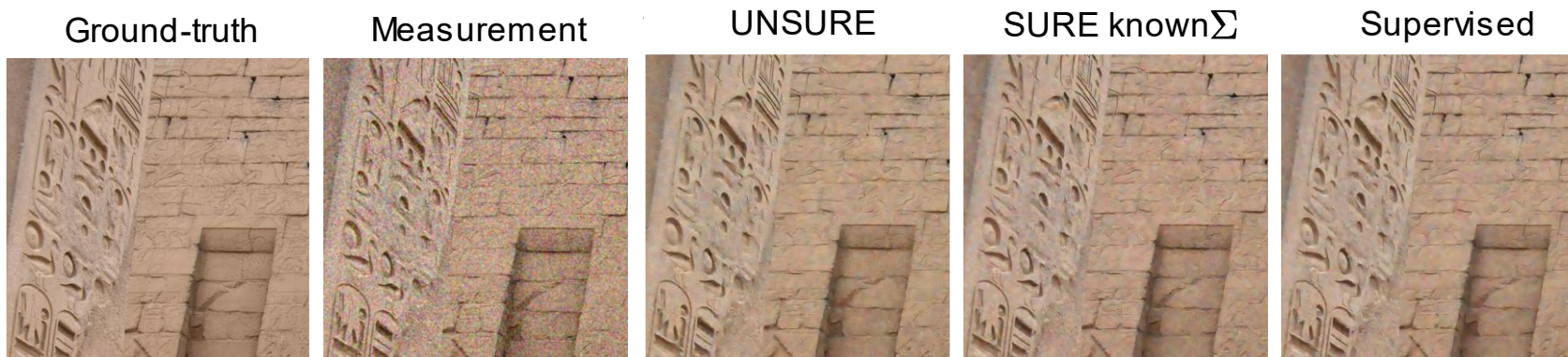
[Tachella & Jacques, *TMLR* 2023]

Correlated Noise Experiments

- DIV2K dataset (320x320 RGB)
- Spatially correlated Gaussian noise 3x3 pixels

UNSURE (unknown Σ)	SURE (known Σ)	Supervised
28.72 ± 1.03	29.77 ± 1.22	29.91 ± 1.26

Kernel size η	1×1	3×3	5×5
PSNR [dB]	23.62	28.72 ± 1.03	27.38 ± 0.88



Efficient SURE

Reconstructed2Reconstructed [Pang et al., 2021], Noisier2Noise [Moran et al., 2020], etc.

Proposition: Let $\mathbf{y} \sim \mathcal{N}(\mathbf{x}, I\sigma^2)$ and define

$$\mathbf{y}_a = \mathbf{y} + \alpha \boldsymbol{\omega}$$

$$\mathbf{y}_b = \mathbf{y} - \boldsymbol{\omega}/\alpha$$

where $\boldsymbol{\omega} \sim \mathcal{N}(\mathbf{0}, I\sigma^2)$ and $\alpha \in \mathbb{R}$, then $\mathbf{y}_a$ and $\mathbf{y}_b$ are **independent** random variables (fixed $\mathbf{x}$).

$$\mathcal{L}_{\text{R2R}}(\mathbf{y}, f, \alpha) = \mathbb{E}_{\boldsymbol{\omega}} || \mathbf{y}_b - f(\mathbf{y}_a) ||^2$$

Generalized Reconstructed2Reconstructed [Monroy, Bacca & Tachella, CVPR 2025]

- **Equivalent to SURE** in the limit: $\lim_{\alpha \rightarrow 0} \mathcal{L}_{\text{R2R}}(\mathbf{y}, f, \alpha) = \mathcal{L}_{\text{SURE}}(\mathbf{y}, f)$
- Extensions to **Poisson, Gamma and Binomial**

Uncertainty Quantification

Can we measure the uncertainty of the reconstructions?

Self-supervised losses can also be used for uncertainty quantification!

- SURE can be used to assess reconstruction error in denoising
- SURE4SURE [Bellec et al., 2021] gives error variance estimates.
- EI loss can be seen as a bootstrapping technique [T. & Pereyra, 2024] with well calibrated uncertainty estimates



Implementation

We parameterize f as a deep neural network, small $\alpha > 0$, $\boldsymbol{\omega} \sim \mathcal{N}(\mathbf{0}, I)$

1) UNSURE: Solve Lagrangian problem, approximating divergence as [Ramani et al., 2007]

$$\text{tr}(\Sigma \frac{\delta f}{\delta \mathbf{y}}) \approx \frac{(\Sigma \boldsymbol{\omega})^\top}{\alpha} (f(\mathbf{y}) - f(\mathbf{y} + \boldsymbol{\omega} \alpha))$$

2) UNSURE via score: Learn score $s(\mathbf{y}) \approx \nabla \log p_{\mathbf{y}}(\mathbf{y})$ via

$$\arg \min_s \mathbb{E}_{\mathbf{y}, \boldsymbol{\omega}} ||\boldsymbol{\omega} - \alpha s(\mathbf{y} + \boldsymbol{\omega} \alpha)||^2$$

and use $f(\mathbf{y}) = \mathbf{y} + \Sigma_{\hat{\boldsymbol{\eta}}} s(\mathbf{y})$ to denoise at test time.